

GROUP THEORY

①

Binary Operation

Let G be non empty set. Then $G \times G = \{(a, b) : a \in G, b \in G\}$.

If $f : G \times G \rightarrow G$, then f is said to be Binary operation on G .

Binary operation denoted by $+$, $\cdot$, $\circ$, $*$ etc.

Closure Property

The $*$ will be a binary operation on G if & only if $a * b \in G$ for all $a, b \in G$ and $a * b$ is unique.

This is said to be the closure property of binary operation & the set G is said to be closed with respect to the binary operation.

Eg. Let there be set of integers $Z = \{\dots -3, -2, -1, 0, 1, 2, 3, \dots\}$.

$$-3, -2 \in Z$$

$$-3 + (-2) = -5 \in Z$$

Solⁿ Binary operation on Z then Z is said to be closed w.r.t the binary operation (+).

Divide operation is not binary operation.

There can be more than 1 binary operation in a single set.

Algebraic Structure

A non-empty together with one or more than one binary operation is called Algebraic Structure

$(\mathbb{Z}, +)$ is Algebraic structure

Eg $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

$(\mathbb{Z}, +, -, \cdot)$ is algebraic structure

Sets

$\mathbb{N}$ = sets of Natural numbers

$\mathbb{Z}$ = set of Integers

$\mathbb{R}$ = set of Real Numbers

$\mathbb{Q}$ = set of Rational Numbers

$\mathbb{C}$ = set of Complex Numbers

Binary operation (composition)
↓

Closure property
↓

Algebraic structure

$$G \times G \rightarrow G$$

$$a, b \in G$$

$$a + b \in G$$

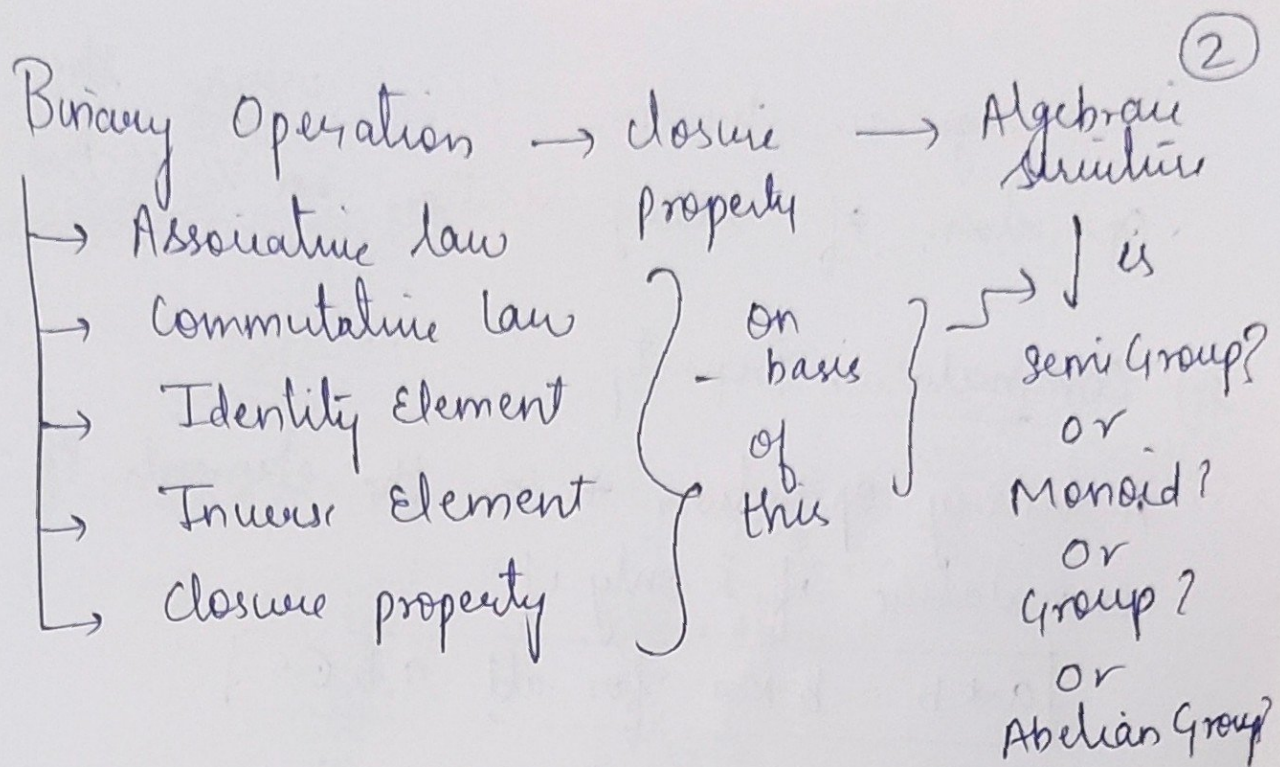
$$\mathbb{Z}$$

$$3, 5 \in \mathbb{Z}$$

$$3 + 5 = 8 \in \mathbb{Z}$$

or

$$3 \cdot 5 = 15 \in \mathbb{Z}$$



Laws / Properties of Binary operation

i) Associative Property

A binary operation $*$ on set S is said to be Associative if & only if

$$a * (b * c) = (a * b) * c \text{ for all } a, b, c \in S$$

Eg Let there be Algebraic Structure $(\mathbb{Z}, +)$ Check whether Associative property holds

Solⁿ $\mathbb{Z} = \{ \dots, -3, -2, -1, 0, 1, 2, 3, \dots \}$

Let $-3, -2, 4 \in \mathbb{Z}$

$$-3 + (-2 + 4) = (-3 + (-2)) + 4$$

$$-3 + (2) = (-3 - 2) + 4$$

$$-1 = -5 + 4$$

$$-1 = -1$$

LHS = RHS so it holds

The algebraic structure $(Z, +)$ where the binary operation of ~~adding~~ addition on Z is Associative.

2) Commutative Property

A binary operation $*$ on the elements of set is commutative if & only if

$$a * b = b * a \text{ for all } a, b \in S$$

Eg let algebraic structure $(\mathbb{Z}, +)$

$$Z = \{0, \dots, -3, -2, -1, 0, 1, 2, 3, \dots, \infty\}$$

Let $-7, 6 \in \mathbb{Z}$

$$\begin{array}{rcl} -7 + 6 & = & 6 + (-7) \\ -1 & = & -1 \end{array}$$

The Algebraic structure $(Z, +)$ where the binary operation of Addition on Z is commutative.

Eg. let there be Algebraic structures $(Z, \cdot)$
 A'' " $(Z, -)$

9)

b) which algebraic structure holds both property & if not holding both property then which property does not hold by Algebraic structure?

21ⁿ

$$\mathbb{Z} = \{\infty, \dots, -3, -2, -1, 0, 1, 2, 3, \dots, \infty\}$$

a) $(\mathbb{Z}, \cdot)$

let $-4, 3, 2 \in \mathbb{Z}$

$$-4 \times (3 \times 2) = (-4 \times 3) \times 2$$

$$-4 \times 6 = -12 \times 2$$

$$-24 = -24$$

Associative

$$-4 \times 3 = 3 \times -4$$

$$-12 = -12$$

commutative

b) $(\mathbb{Z}, -)$

let $-8, -12, 6 \in \mathbb{Z}$

~~$$-8 \times (12 \times 6) = (-8 \times 12) \times 6$$~~

$$-8 - (-12 - 6) = (-8 - (-12)) - 6$$

$$-8 - (-18) = (-8 + 12) - 6$$

$$-8 + 18 = 4 - 6$$

$$10 = -2$$

$$-8 - (-12) = -12 - (-8)$$

$$-8 + 12 = -12 + 8$$

$$4 = -4$$

Identity Element

An element in a set S is called an Identity element w.r.t Binary operation $*$ if for any element a in S

$$\boxed{a * e = e * a = a}$$

↓
Identity element

Different e for different binary operations

Eg $\mathbb{Z} = \{ \dots -3, -2, -1, 0, 1, 2, 3 \dots \}$

a) $(\mathbb{Z}, +)$ check for Identity element

Solⁿ $\mathbb{Z}, + : \overset{\text{let } 1}{-5} + 0 = -5 = 0 + (-5)$

↓
Identity element

So 0 is the identity element w.r.t Binary operation $+$.

b) $(\mathbb{Z}, \cdot)$

Solⁿ $9 \times 1 = 9 = 1 \times 9$

↓
Identity element

So 1 is Identity element w.r.t Binary operation $\cdot$.

$$a * e = e * a = a$$

$\downarrow$ Right Identity $\downarrow$ Left Identity

Inverse Element

Consider a set S having the identity element e w.r.t binary operation $*$. If corresponding to each element $a \in S$ there exists an element $b \in S$ such that

$$a * b = b * a = e$$

Then b is said to be the Inverse of a & is usually denoted by a^{-1} .

Eg $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

$(\mathbb{Z}, +)$:

$$-10 + \boxed{10} = 0 = \boxed{10} + (-10)$$

$$25 + \boxed{-25} = 0 = \boxed{-25} + 25$$

$\downarrow$
 $e = 0$

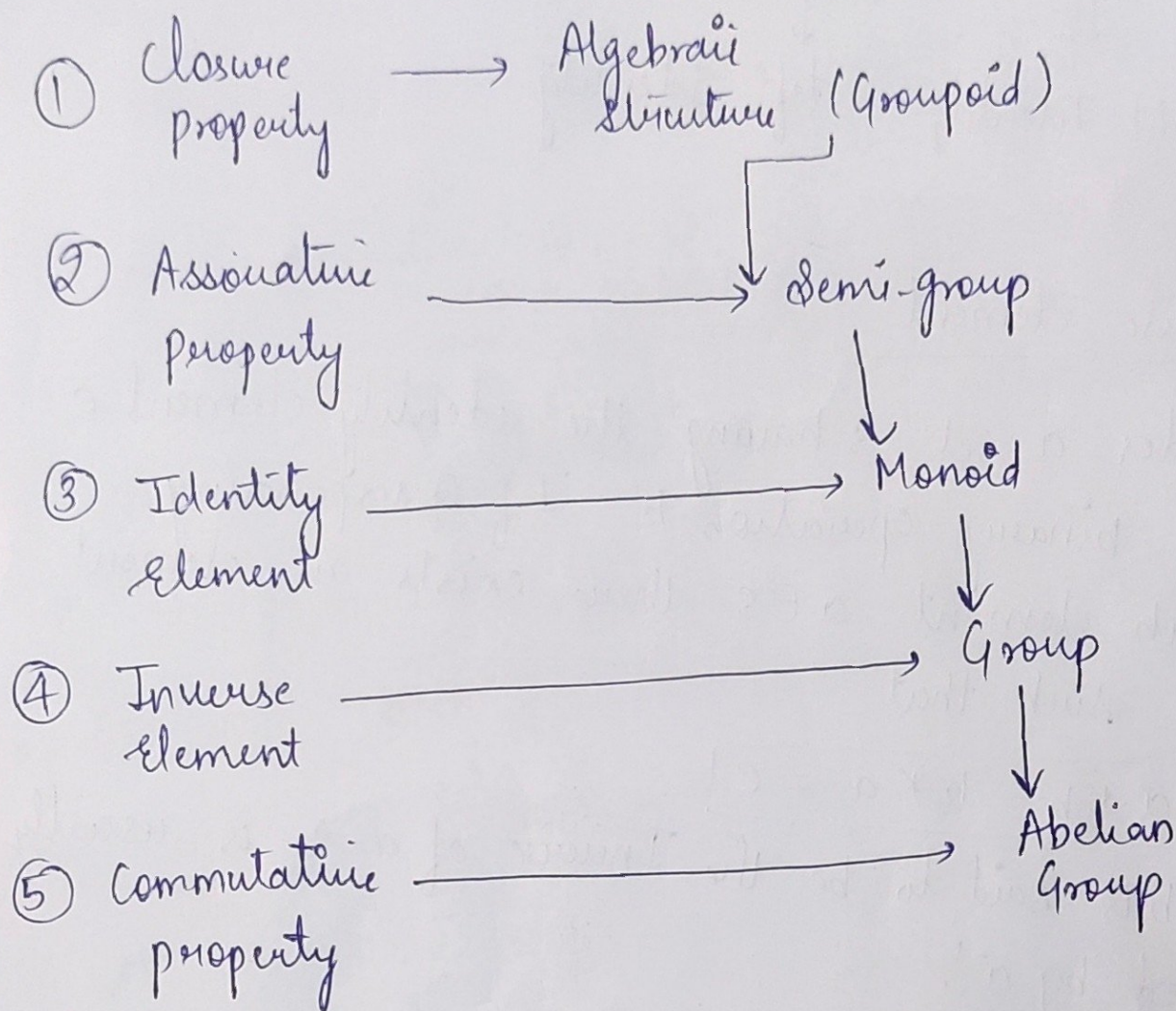
$(\mathbb{Z}, \cdot)$

$\downarrow$
 $e = 1$

$$15 \times \left(\frac{1}{15}\right) = 1 = \left(\frac{1}{15}\right) \times 15$$

No Inverse element for binary operation $(\cdot)$ as the a^{-1} must belong to $\mathbb{Z}$.

Group



Group

Let $(G, *)$ be an algebraic structure where $*$ is a binary operation, then $(G, *)$ is called a Group under this operation if the following conditions are satisfied.

① Closure property

The binary operation $*$ is called closed operation if

$$a * b \in G \quad \forall a, b \in G$$

② Associative property

The binary operation $*$ is an associative property if

$$(a * b) * c = a * (b * c) \quad \forall a, b, c \in G$$

③ Identity Element

There exists an element $e \in G$ such that

$$a * e = e * a = a$$

④ Inverse element

for each $a \in G$, there exists an element $a^{-1} \in G$ such that $a \times a^{-1} = a^{-1} \times a = e$

Then a^{-1} is called Inverse of a in G .

Eg let $(\mathbb{R}, +)$

1) closure $\Rightarrow 10 + 15 = 25 \in \mathbb{R}$

2) Associative

$$3 + (10 + 2) = (3 + 10) + 2$$
$$15 = 15$$

3) Identity element

$$5 + 0 = 5$$

$$-10 + 0 = -10$$

$0 \rightarrow$ Identity element

4) Inverse element - Inverse element of every element

$$23 + (-23) = 0 \rightarrow [\text{identity}]$$

$$-29 + 29 = 0 \rightarrow$$

$(\mathbb{R}, +)$ is a Group Ans

Q Prove that the fourth root of unity
1, -1, i, -i form a ^{multiplicative} group

Solⁿ Let $G = \{1, -1, i, -i\}$

- Composition table

$\times$	1	-1	i	-i
1	①	-1	i	-i
-1	-1	①	-i	i
i	i	-i	-1	①
-i	-i	i	①	-1

$$\because i^2 = -1$$

$$\because -i^2 = 1$$

$$i = \sqrt{-1} \times \sqrt{-1}$$

$$= -1$$

$$-i^2 = -(-1)$$

$$= 1$$

check for Group

1) closure property $\rightarrow$ Each element in the Table belongs to set G. It holds closure property or G is closed w.r.t multiplication

2) Associative property 1, i, -i

$$1 \times (i \times -i) = (1 \times i) \times -i$$

$$1 \times (1) = i \times -i$$

$$1 = 1$$

Holds this property

3) Identity Element

$1 \in G$ is identity element as

$$1 \times a = a \times 1 = a \quad \boxed{e = 1}$$

4) Inverse Element

$$1 \times 1 = 1$$

$$-1 \times -1 = 1$$

$$i \times -i = 1$$

$$-i \times i = 1$$

Inverse of 1 is 1

" " -1 is -1

" " i is -i

" " -i is i

and all those belongs to G .

Hence it follows that G is a group.

Note $(G, +) \rightarrow$ Additive Group
 $(G, \bullet) \rightarrow$ Multiplicative Group

Q $(\mathbb{R}, \cdot)$ is a Group or not?

Q $(\mathbb{R}^*, +)$ is Group or not?

$\mathbb{R}^*$ means set of non-zero real numbers

Theorem of Identity element

The Identity element (if it exists) of any algebraic structure is unique

or

Identity element is unique in a Group

Proof Let e & e' be 2 identity elements of the group $(G, *)$, Hence $e, e' \in G$.

Now e is an identity element

$$e * e' = e' \quad \text{--- (1)}$$

e' is an identity element

$$e * e' = e \quad \text{--- (2)}$$

$$\Rightarrow e = e'$$

Thus the identity element is unique in Group $(G, *)$

Theorem : Inverse of each element in a Group is Unique

Proof :-

Let $(G, *)$ be Group

e is Identity element of Group

Let the inverse of each element be not unique

If $a \in G$ is an element of G then let

b & c are 2 inverses of a

→ If b is considered an inverse of a then
 $a \times b = b \times a = e$ — (1)

→ If c is an inverse of a then
 $a \times c = c \times a = e$ — (2)

Now from Associative property

$$\begin{aligned}(b * a) * c &= b * (a * c) \\ e * c &= b * e\end{aligned}$$

$$c = b \quad \left\{ \begin{array}{l} \text{by Identity element} \\ \because e * c = c \\ b * e = b \end{array} \right.$$

Both inverses of element a

are same

Hence there cannot be 2 different inverse of an element
∴ Inverse for each element in a Group is Unique.

Abelian Group

A Group G is said to be Abelian if the commutative holds
ie $a * b = b * a$ for all $a, b \in G$

Eg Prepare the composition table for multiplication on the element in the set $A = \{1, \omega, \omega^2\}$ where ω is cube root of unity. Show that $(A, *)$ is an Abelian Group.

Solⁿ

$\omega \rightarrow$ cube root of unity

$$\omega^3 = 1$$

composition table

X	1	ω	ω^2
1	①	ω	ω^2
ω	ω	ω^2	①
ω^2	ω^2	①	ω

$$\therefore \omega^3 = 1 \quad [\omega \times \omega^2 = \omega^3]$$

$$\omega^2 \times \omega^2 = \omega^4$$

$$\omega^3 \times \omega$$

$$1 \times \omega$$

$$= \omega$$

1) Closure property

$$a * b \in G$$

$a * b \in A$ i.e. closure property satisfied

2) Associative property

$$a * (b * c) = (a * b) * c \in A$$

$$\omega * (1 \times \omega^2) = (\omega \times 1) \times \omega^2$$

$$\omega \times \omega^2 = \omega \times \omega^2$$

$$\omega^3 = \omega^3$$

$$1 = 1$$

Holds this property

3) Identity element

$$a * e = e * a = a$$

$$1 \times \overset{1}{\underset{\uparrow}{1}} = \overset{1}{\underset{\uparrow}{1}} \times 1 = 1$$

$$\omega \times \overset{1}{\underset{\uparrow}{1}} = \overset{1}{\underset{\uparrow}{1}} \times \omega = \omega$$

$$\omega^2 \times \overset{1}{\underset{\uparrow}{1}} = \overset{1}{\underset{\uparrow}{1}} \times \omega^2 = \omega^2$$

$$e = 1$$

4) Inverse element

$$a \times a^{-1} = a^{-1} \times a = e$$

$$1 \times 1 = 1 \times 1 = 1$$

$$\omega \times \omega^2 = \omega^2 \times \omega = 1$$

$$\omega^2 \times \omega = \omega \times \omega^2 = 1$$

$\{1, \omega^2, \omega\}$
are inverse
elements

⑤ Commutative property

$$a * b = b * a \quad \forall a, b \in A$$

$$\left. \begin{array}{l} w \times 1 = 1 \times w \\ w = w \end{array} \right\} \begin{array}{l} w \times w^2 = w^2 \times w \\ 1 = 1 \end{array}$$

Holds this property

$(A, \cdot)$ is an Abelian Group

Semi-Group

An algebraic structure $(S, *)$ is called Semi group if following condⁿ are satisfied

1) closure property

$$a * b \in S \quad \forall a, b \in S$$

2) Associative Property

$$a * (b * c) = (a * b) * c \quad \forall a, b, c \in S$$

Monoid

An algebraic structure $(S, *)$ is called monoid if the following condⁿ satisfied

- 1) closure property $(a*b)$
- 2) Associative property $a*(b*c) = (a*b)*c \quad \forall a, b, c \in S$
- 3) Identity element $a*e = e*a = a \quad \forall a \in S$

or

A semigroup holding an identity element then is called Monoid

Eg Determine whether the set together with the binary operation is a semigroup or monoid

a) $\mathbb{Z}^+$, where $a*b = \max(a, b) \quad \forall a, b \in \mathbb{Z}^+$

Solⁿ $\mathbb{Z}^+ \rightarrow$ set of positive integers

$$(a*b) = \max(a, b)$$

① closure property

$$100, 200 \in \mathbb{Z}^+$$

$$(100 * 200) = 200 \in \mathbb{Z}^+ \\ \hookrightarrow \max(a, b)$$

2) Associative property

50, 100, 500

$$\begin{aligned} 50 * (100 * 500) &= (50 * 100) * 500 \\ 50 * \underbrace{(100 * 500)}_{\text{max}} &= \underbrace{(50 * 100)}_{\text{max}} * 500 \\ \underbrace{500}_{\text{max}} &= 500 \end{aligned}$$

Holds this property

2 properties holds so it's a semi group

3) Identity element

$$\begin{array}{c} \xrightarrow{\text{max}} \\ 50 * 1 = 50 \end{array}$$

$$\begin{array}{c} 250 * 1 = 250 \\ \xrightarrow{\text{max}} \end{array}$$

Identity element $e = 1$

$(\mathbb{Z}^+, *)$ is a monoid

Sub Group

Let $(G, *)$ be a group & H is a subset of G .
 $(H, *)$ is said to be subgroup of G if $(H, *)$ is also group by itself.

eg Let $G = \{1, -1, i, -i\}$ is a multiplicative group &
 $H = \{1, -1\}$ where $H \subseteq G$, then show that H is subgroup of G .

Solⁿ composition table

$\times$	1	-1
1	1	-1
-1	-1	1

① closure property
 $a * b$
 1×-1

② Associative property

$$\begin{aligned} a * (b * c) &= (a * b) * c \\ 1 * (1 * -1) &= (1 * 1) * -1 \\ -1 &= -1 \end{aligned}$$

③ Identity element

$$a * e = a \quad e = 1$$

$$1 * 1 = 1$$

④ Inverse element for each & every element in composition table

$$a * a^{-1} = e$$

$$a^{\circ} = a^{-1}$$

$$1 = 1$$

$$1 \times 1 = 1$$

$$-1 = -1$$

H is a group Hence H is also subgroup of G

Note

① Every set is a subset of itself. Therefore if G is a group then itself is a subgroup of G .

② Two subgroups $(G, *)$ and $(\{e\}, *)$ of the group $(G, *)$ are called Improper or Trivial subgroups, others of are called Proper or Nontrivial subgroups.

Other Examples

① The additive group of even integers is a subgroup of the additive group of all integers.

② The set $\mathbb{Q}^+$ of all non zero positive rational numbers is a subgroup of multiplicative group $\mathbb{Q}^*$ of all non-zero rational numbers.

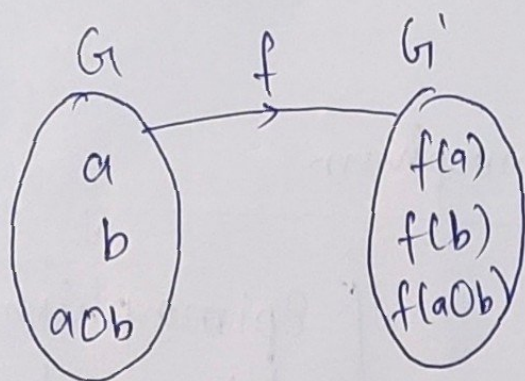
Homomorphism & Isomorphism

$\Rightarrow$ Homomorphism of a Group

Let (G, O) and (G', O') are two groups. Then mapping $f: G \rightarrow G'$ is said to be Homomorphism if

$$f(aOb) = f(a) O' f(b)$$

where a & $b \in G$ & $f(a), f(b) \in G'$
i.e. preserve composition or preserve operation



$\Rightarrow$ Isomorphism of a Group

Let (G, O) & (G', O') are two groups then a mapping $f: G \rightarrow G'$ is called Isomorphism if

a) f is one-one
 $\forall a, b \in G$

$$\begin{aligned} \Rightarrow f(a) &= f(b) \\ \Rightarrow a &= b \end{aligned}$$

ii) f is onto

$$\forall b \in G$$

$\Rightarrow \exists$ an element $a \in G$

$$\text{such that } f(a) = b$$

iii) f is preserve composition or f is preserve operation or f is Homomorphism

$$\forall a, b \in G$$

$$f(a \circ b) = f(a) \circ' f(b)$$

where $a, b \in G$

$$f(a), f(b) \in G'$$

Homomorphism



Endomorphism

if homomorphism

$$f: G \rightarrow G$$

$$(G, \circ) \rightarrow (G, \circ)$$

$$f(a \circ b) = f(a) \circ' f(b)$$

Epimorphism/
Ontohomomorphism

if homomorphism

$$f: G \rightarrow G' \text{ is onto}$$

eg $(G, +) \rightarrow G$ is set of real no

$(G', \times) \rightarrow$ set of non zero real no.

$$f(x) = 2^x \quad \forall x \in G.$$

check whether $f(x) : G \rightarrow G'$ is homomorphism

Soln let $a, b \in G$ then $f(a) = 2^a, f(b) = 2^b$

$$f(a+b) = 2^{a+b} = 2^a \cdot 2^b = f(a) \times f(b)$$

$$\boxed{f(a+b) = f(a) \times f(b) \quad \forall a, b \in G}$$

for Endomorphism

$$G \rightarrow G$$

Not Endomorphism
as sets are different

for Epimorphism

$f : G \rightarrow G'$ is onto

$$f(x) = 2^x$$

$$y = 2^x \quad \begin{matrix} y \in G' \\ x \in G \end{matrix}$$

$f(x)$ is not onto

$$\rightarrow \text{let } x = 0 \quad 2^0 = 1$$

$$y = 1 \quad x = 0$$

$$\rightarrow x = -2 \quad -2$$

$$y = 2$$

$$y = \frac{1}{2^2} = \frac{1}{4}$$

eg ① $(G, +)$ Set of Real No
 $(G', \times)$ $G' \rightarrow$ Set of Non zero +ve Real No

$$f(x) = 2^x \quad \forall x \in G$$

check $f(x)$ for Isomorphism

$$f: G \rightarrow G'$$

solⁿ ① f is one-one

Let $x_1, x_2 \in G$

$$\text{ie } f(x_1) = 2^{x_1}$$

$$f(x_2) = 2^{x_2}$$

$$f(x_1) = f(x_2)$$

$$2^{x_1} = 2^{x_2}$$

$$\boxed{x_1 = x_2}$$

$f: G \rightarrow G'$ is one-one

② f is onto

$$\text{let } y = f(x) = 2^x$$

$$y = 2^x$$

$$\log_2 y = \log_2 2^x = x$$

$$\log_2 2^x = x \log_2 2 = x$$

f is onto